Vel Tech Group of Institutions

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2028_NeoColab_Competitive Coding II_L8

VTU_Week 1_COD_SB

Attempt : 1
Total Mark : 50
Marks Obtained : 50

Section 1 : Coding

1. Problem Statement:

A software company is developing a mathematical tool for students to help them understand factorization and modular arithmetic. As part of the tool, they need a function that calculates the product of all positive divisors (factors) of a given integer, with the result taken modulo 1000000007. The tool is intended to assist in solving mathematical problems and providing insights into number theory.

Your task is to implement a function that computes the product of all factors of a given integer n, and then display the result modulo 1000000007.

Examples :

Input: 12

Output: 1728

1 * 2 * 3 * 4 * 6 * 12 = 1728

Input Format

The input consists of a single integer n.

Output Format

The output displays a single integer which is the product of all factors of n modulo 1000000007.

Refer to the sample output for better understanding.

Sample Test Case

Input: 12

Output: 1728

Answer

```
// You are using GCC
#include<stdio.h>

int main(){
   int n;
   long long product=1;
   const long long MOD=1000000007;
   scanf("%d",&n);
   for(int i=1;i<=n;i++){
      if(n%i==0){
         product=(product*i)%MOD;
      }
   }
   printf("%d",product);
   return 0;
}
```

Status : Correct ***Marks : 10/10***

2. Problem Statement

Pradeep Narwal is analyzing multiple numerical ranges to identify prime numbers within each range. He is given several ranges defined by a starting number A and an ending number B. Using the Sieve of Eratosthenes, Pradeep wants to efficiently determine all prime numbers that lie within each range.

Write a program to ensure that all prime numbers between A and B (inclusive) are printed for every given range.

Input Format

The first line of input consists of an integer Q, representing the number of ranges.

The next Q lines each contain two integers A and B , representing the start and end of each range.

Output Format

For each range, print all prime numbers between A and B (inclusive) in a single line.

Prime numbers must be printed in ascending order, separated by a single space.

If a range contains no prime numbers, print an empty line.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2
1 5
10 15

Output: 2 3 5
11 13

Answer

// You are using GCC

```
#include <stdio.h>
#include <stdbool.h>

int main() {
    int Q;
    scanf("%d", &Q);

    int A[100], B[100];
    int max = 0;

    for(int i = 0; i < Q; i++) {
        scanf("%d %d", &A[i], &B[i]);
        if(B[i] > max) max = B[i];
    }

    bool prime[max + 1];
    for(int i = 0; i <= max; i++) prime[i] = true;

    if(max >= 0) prime[0] = false;
    if(max >= 1) prime[1] = false;

    for(int i = 2; i * i <= max; i++)
        if(prime[i])
            for(int j = i * i; j <= max; j += i)
                prime[j] = false;

    for(int i = 0; i < Q; i++) {
        for(int j = A[i]; j <= B[i]; j++)
            if(prime[j]) printf("%d ", j);
        printf("\n");
    }

    return 0;
}
```

Status : Correct ***Marks : 10/10***

3. Problem Statement

Joanna is developing a utility tool for a math club's quiz event. One of the rounds involves finding the greatest common divisor (GCD) of a list of numbers. Instead of computing it manually for each pair, she wants to automate the task using code.

She decides to use Euclid’s Algorithm, a well-known and efficient method to compute the GCD of two numbers. For a list of numbers, she will apply the algorithm iteratively across all elements in the array.

Help Joanna implement a program that takes an array of positive integers and computes the GCD of the entire array using Euclid’s Algorithm.

Make sure the program also handles special cases:

If the array is empty, return -1.If the array has only one number, return that number as the GCD.

Example

Input:

3

12 18 24

Output:

6

Explanation:

GCD(12, 18) = 6GCD(6, 24) = 6Final result = 6

Input Format

The first line of input contains an integer n, representing the number of elements in the array.

The second line contains n space-separated positive integers, each representing an element of the array.

Output Format

The output consists of a single line:

- If the array contains two or more elements, print a single integer representing the GCD of all array elements.
- If the array contains only one element, print that element as the GCD.
- If the array is empty, print -1 as a convention to indicate undefined GCD.

Refer to the sample outputs for the formatting specifications.

Sample Test Case

Input: 2
8 12

Output: 4

Answer

```
#include <stdio.h>

int gcd(int a, int b) {
   while (b != 0) {
      int temp = b;
      b = a % b;
      a = temp;
   }
   return a;
}

int main() {
   int n;
   scanf("%d", &n);

   if (n == 0) {
      printf("-1");
      return 0;
   }

   int num, result;

   scanf("%d", &result);   // first number

   for (int i = 1; i < n; i++) {
```

```
        scanf("%d", &num);
        result = gcd(result, num);
    }

    printf("%d", result);

    return 0;
}
```

Status : Correct ***Marks : 10/10***

4. Problem Statement

Arjun is working on a project where he needs to analyze a series of integers within a specified range [A, B]. The project involves identifying certain numbers that are not divisible by any smaller positive integers other than 1 and themselves. These numbers are crucial for the next step of his analysis, as they have unique mathematical properties.

Arjun has been asked to implement an efficient way to find all such numbers in a given range. Given that the range can be large, he needs to ensure the solution is optimized.

Can you help him implement this task using the Simple Sieve of Eratosthenes?

Input Format

The first line of input consists of an integer A, representing the starting range.

The second line of input consists of an integer B, representing the ending range.

Output Format

The output prints a space-separated list of numbers that are not divisible by any smaller positive integers other than 1 and themselves. within the range [A, B].

Refer to the sample output for formatting specifications

Sample Test Case

Input: 2
10

Output: 2 3 5 7

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdbool.h>

int main() {
   int A, B;
   scanf("%d", &A);
   scanf("%d", &B);

   bool prime[B + 1];

   for (int i = 0; i <= B; i++)
      prime[i] = true;

   if (B >= 0) prime[0] = false;
   if (B >= 1) prime[1] = false;

   for (int i = 2; i * i <= B; i++) {
      if (prime[i]) {
         for (int j = i * i; j <= B; j += i)
            prime[j] = false;
      }
   }

   int first = 1;  // to avoid extra space

   for (int i = A; i <= B; i++) {
      if (prime[i]) {
         if (!first)
            printf(" ");
         printf("%d", i);
         first = 0;
      }
   }
```

```
    return 0;
}
```

Status : Correct **_Marks : 10/10_**

5. Problem Statement

Raychel is working on a calendar synchronization tool where she needs to determine how often two repeating tasks will coincide. Each task occurs after a fixed number of days, and she must find the least number of days after which both tasks align again — in other words, the least common multiple (LCM) of the two intervals.

Raychel knows that the most efficient way to compute the LCM is by first calculating the greatest common divisor (GCD) using Euclid's Algorithm, and then using that result to derive the LCM. She also wants to make sure the program works for large inputs without running into overflow errors.

Help Raychel implement a program that calculates the LCM of two non-negative integers by first using Euclid's Algorithm to compute their GCD, and then deriving the LCM from it.

Example

Input:

15 20

Explanation:

GCD(15, 20) = 5LCM = (15 / 5) × 20 = 3 × 20 = 60

Output:

60

Input Format

The input consists of a single line containing two space-separated integers 'a' and 'b', representing the two input numbers.

Output Format

The output consists of a single line:

- If both 'a' and 'b' are non-zero, print a single integer — the least number that is divisible by both.
- If either 'a' or b is '0', print 0 as the result since no common multiple exists in such cases.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 15 20

Output: 60

Answer

```
// You are using GCC
#include <stdio.h>

long long gcd(long long a, long long b) {
    while (b != 0) {
        long long temp = b;
        b = a % b;
        a = temp;
    }
    return a;
}

int main() {
    long long a, b;
    scanf("%lld %lld", &a, &b);

    if (a == 0 || b == 0) {
        printf("0");
        return 0;
    }
```

```
    long long g = gcd(a, b);
    long long lcm = (a / g) * b;

    printf("%lld", lcm);

    return 0;
}
```

Status : Correct ***Marks : 10/10***